Holcombe Department of Electrical and Computer Engineering
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Lab Report
Experiment 3 Computer Analysis

ECE 008

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1) Abstract

In this third experiment, the computer software LT-Spice is used to build circuits. These circuits were simulated, and the relevant information was noted. Next, the value of voltage drops, and currents are solved by hand. This is to verify consistent results. The purpose of this experiment is to showcase the value that computer simulated circuits can add to electrical engineering, as well as learn how to utilize the software correctly and reliably.

2) Introduction

The main goal of the experiment is to show proficiency, and the importance of LT-Spice. This is achieved by recreating a parallel and series circuit, and simulating values. This is extremely applicable to the real world, and more importantly, the real work of engineers. Not all calculations are done by hand, so having knowledge and experience with simulation software will prove to be valuable assets. Simulation software like LT-Spice, as well as Multisim and others are great learning tools that will also be applied in the future.

3) Theory and Procedure

Circuit Analysis requires understanding and application of the fundamental laws of circuits. The first being Ohm's Law (1).

$$V = IR \tag{1}$$

Ohm's Law states that the voltage at a point in the circuit is equal to the current multiplied by the resistance. The other two laws are both Kirchhoff's Laws. The Kirchhoff Current Law (2) states that the sum of currents entering and leaving a node equal each other.

$$\Sigma I_{in} = \Sigma I_{out} \tag{2}$$

The final fundamental law, and Kirchhoff's second law, is Kirchhoff's Voltage Law (3). This law states that the sum of voltage rises and drops around a circuit equal zero.

$$\Sigma V_{Loop} = 0 \tag{3}$$

With these three laws, the circuits used in the experiment can be solved by hand through applying all three of these laws to both circuits.

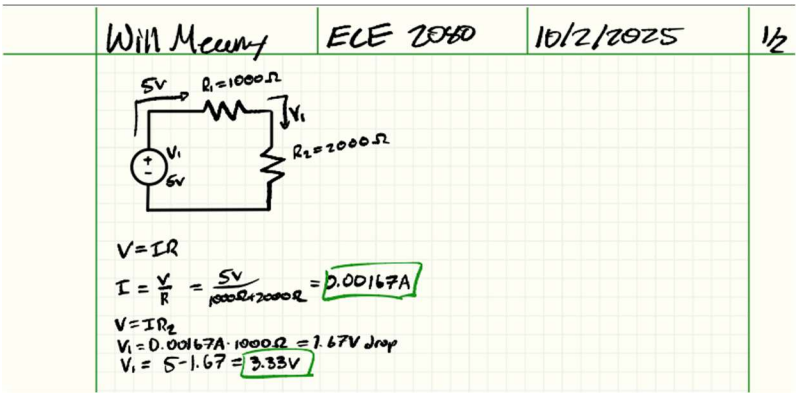
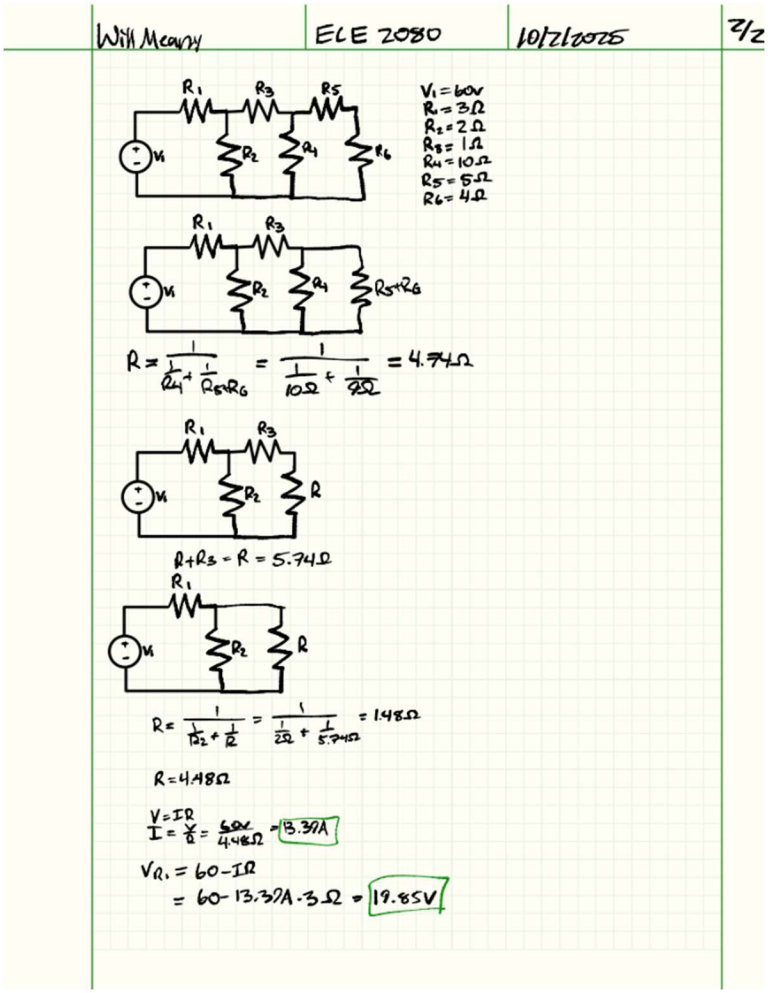


Figure 1: Analysis of Circuit 1 by hand



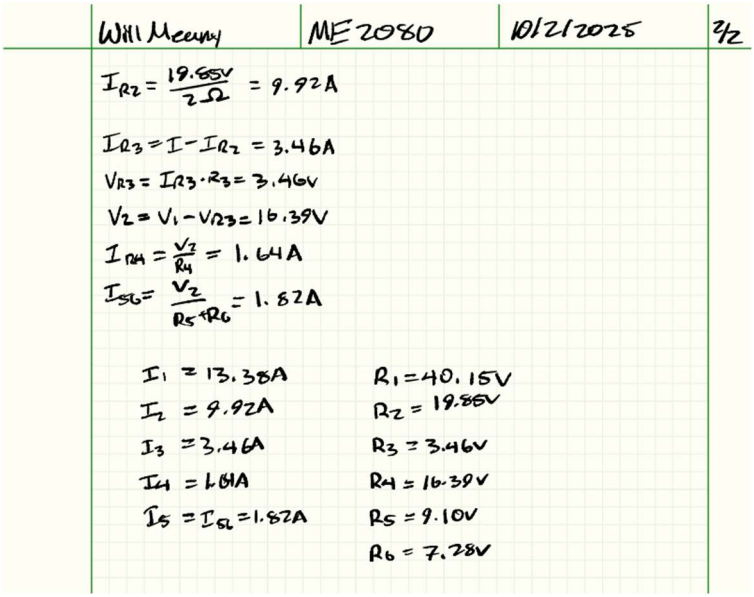


Figure 2: Analysis of Circuit 2 by hand

I. The software LT-Spice is downloaded and installed onto the laptop.
II. After opening the software, a paper with the circuits on it was handed out.
III. The first circuit, a series, is recreated in LT-Spice.
IV. A simulation was run, and results are documented. The circuit and results are visible in Appendix B: Figure 1.
V. The second circuit, a parallel circuit is now recreated in LT-Spice
VI. The second circuit is simulated, and results are documented in Appendix B: Figure 2.
VII. Following these results, hand calculations re made using formulas (1,2,3) to both confirm the results of LT-Spice, as well as circuit analysis skills.

4) Discussion of Results

The results of the first simulation (Appendix B: Figure 3) are the current across the resistors, which is the same due to the first circuit being a series circuit, and the voltage across the resistors. The current is shown as negative, which is not possible. This is due to the resistors being directional in the simulation software, contrary to real-world application. This means that the result of -0.00167 Amps can be assumed as 0.00167 Amps. These values seem reasonable at a glance, the very low current is explained by the relatively high resistance to the low voltage. These results are also proven by hand calculations above.

The results of the second simulation (Appendix B: Figure 4) yield much more information than the first with four different voltage drops over resistors, as well as six values of current. Similarly to the first one, the negative values of current can be assumed positive. The nature of parallel circuits being more difficult to solve makes LT-Splice greatly useful here, and a great time saver. Unlike the first circuit, the hand calculations are not an exact match, but they are close such that the error could be assumed as rounding. (Figure 2).

The results of the experiment are close to both hand calculations. LT-Spice can be assumed to be more accurate considering it keeps track of more decimal places and does not make calculation errors. The goal of this experiment is to showcase how valuable knowing how to utilize simulation software, and this could be seen through the second circuit. Though it took some time to make in LT-Spice, it took less time to replicate the circuit and simulate it, than to solve it by hand. In addition to the increase in speed, if the circuit is made properly, the information will be more accurate. The goal of this lab is to showcase the advantages and disadvantages of LT-Spice. The advantages include precision and speed. These are balanced with the disadvantages of human error. The circuit can either be created improperly, or the results can be drawn incorrectly.

5) Conclusions

The experiment yielded more than numerical results. It was an introduction to the software LT-Splice, as well as its strengths and weaknesses. Knowing this is crucial for proper usage of the software, and attempting to properly apply the software to higher level Electrical Engineering work could lead to incorrect results, as well as confusion. LT-Spice is an incredible tool for engineers, and through combining what has been learned with new information, it can be used to strengthen the skills of engineers.

References

[1] A. L. Duke and D. McAuliff, *Electrical Engineering Laboratory I*.

[2] Analog Devices, "LTspice Information Center | Analog Devices," *Analog.com*, 2024. https://www.analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html

[3] IEEE, "IEEE SA - The IEEE Standards Association - Home," *Ieee.org*, 2019. https://standards.ieee.org/

Appendix A: Technical Standards

IEEE/ANSI 315A-1986, NFPA 70e

Appendix B: Flow-Chart(s) and/or State Diagrams(s) and/or Computer Code

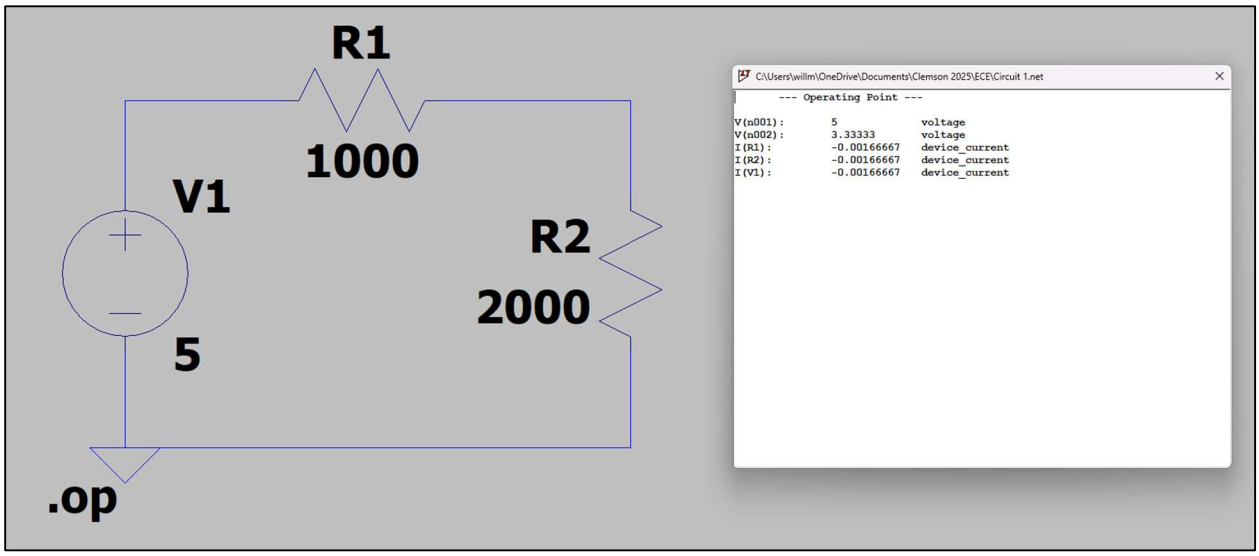


Figure 3: LT-Spice Circuit 1: Series Circuit

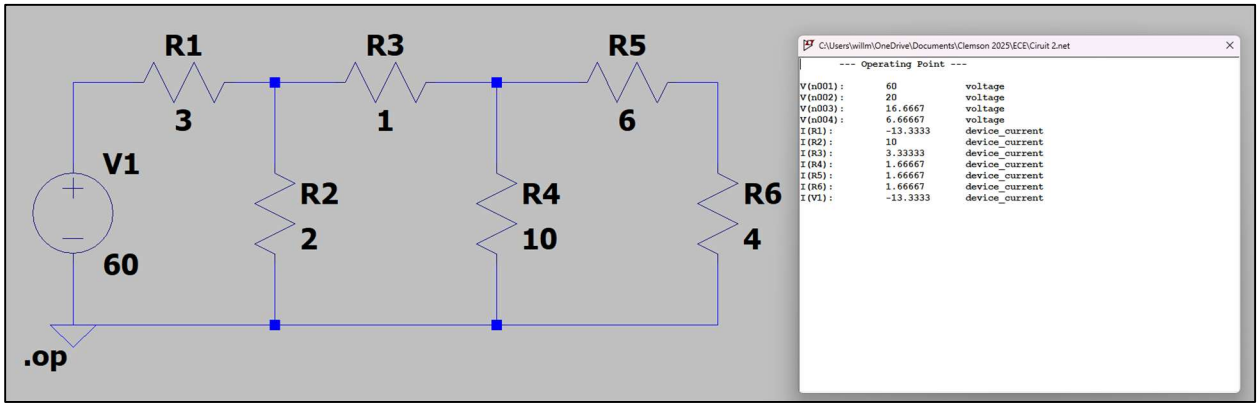


Figure 4: LT-Spice Circuit 2: Parallel Circuit